

Certified analytical bounds for hypersonic reachable sets

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Abstract

This is a proposal in nonlinear analysis and dynamical systems. The object is a singularly perturbed family of differential inclusions in which compactness fails along the time axis: as the parameter vanishes, the motion that carries the important state transitions occupies a vanishing time, so it disappears from the ordinary limits of the family while continuing to decide where the trajectories go. The remedy we propose is a *profile decomposition* — a low-dimensional *skeleton*, finitely many concentrating *bubbles*, and a *residual* bounded in a stated metric — and the question is whether that decomposition is a theorem about solution sets rather than a description of them.

We state the question as a hierarchy. *Theorem A*: every sequence of trajectories splits, along a subsequence, into a skeleton, finitely many rescaled profiles above any fixed size, and a remainder. *Theorem B*: the profiles induce a set-valued jump map. *Theorem C*: the solution set of the family is contained in that of the skeleton-and-jump system, thickened by an explicit ball. This document proves Theorem C, with every constant written down, for a model whose potential is asymmetric in the one way that produces concentration, and whose profiles are counted *a priori* by a Lyapunov functional rather than extracted by an iterated dichotomy. That is where we expect the contribution to the analysis to lie: the classical concentration-compactness argument yields the structure by contradiction and produces no constant, whereas a dissipation functional yields the same structure quantitatively — here as an explicit bound on the number of profiles above a given size, and on the amplification they compound.

Hypersonic atmospheric entry is the example that supplies the problem and the one against which the result is to be measured; it is not the subject. [Section 7](#) states the correspondence and what the full system adds. Only the theorem of [Section 4](#) is claimed as proved.

1 The question, reduced

The object we want enclosed is the solution set of a family of differential inclusions: every state reachable under every admissible control and every admissible selection from the field. Two things make it hard to enclose. The first is its dimension, which puts the exact methods out of reach. The second is the nature of the enclosure: a bound is worth having only if it comes with a proof, and the proof is to be a pair of functions whose superlevel set contains the solution set — exhibited and checked, rather than produced by a solver and inheriting its accuracy ([Section 6](#)). This document is narrower than the programme and prior to it: it takes the one analytical difficulty on which the programme turns, strips it of the application, and proves a theorem about it.

The difficulty is a singular limit of a kind that defeats the ordinary compactness arguments. The example supplies the parameter: for atmospheric entry it is the ratio $\varepsilon = H_s/R_e$ of the atmospheric scale height to the planetary radius, whose square root is the ratio of the period of the oscillation about equilibrium to the slow time of the flight. As $\varepsilon \rightarrow 0$ the motion separates into arcs on which the forcing is not felt and *passes* in which it is felt strongly for a vanishing time. A pass becomes instantaneous and keeps its effect, so in the limit a trajectory is a sequence of arcs joined by jumps — and yet the natural limiting objects do not see the

jumps, because they weigh an interval of time by its length and a pass has none. A solution set is a support and not an integral, so what has no mass still counts.

This is the structural phenomenon of concentration, and the language for it is a profile decomposition: a low-dimensional *skeleton*, which is what survives the limit; one *bubble* per pass, the pass rescaled to its own width, whose net effect on the surviving coordinates is a jump; and a *residual*, which is what the skeleton and the finitely many bubbles do not capture. In the theory of nonlinear dispersive equations such a decomposition is obtained by iterating the dichotomy of the concentration-compactness principle [1, 2], and what it delivers is structure: it says which pieces can occur and that nothing else can, by contradiction, and it produces no constant. The authors of the wave-equation theory are explicit that the uniform bound following from their decomposition carries a constant they cannot describe [2, Corollary 2].

Our proposal is that here the structure can be had *quantitatively*, because the family carries a Lyapunov functional. The dissipation along a trajectory is monotone and its total variation is bounded uniformly in the parameter, so the number of profiles above any threshold is bounded in advance rather than extracted by a contradiction argument, and the same functional bounds the time the motion may spend concentrating and the accumulated growth of the comparison. We therefore state the question as the hierarchy of the abstract — the decomposition, the jump map it induces, and the inclusion for solution sets — and this document proves the third for a model that keeps the concentration and discards everything else.

Remark 1.1 (What the small parameter is, and what it is not). It is tempting to write the reduced problem as a slow state driven by a fast one, $\dot{z} = g(z) + h(y)$ with $\varepsilon \dot{y} = q(y, z)$ and h bounded. That form does not contain the difficulty: if the excursion in y lasts a time of order ε and the forcing is of order one, its contribution to z is of order ε , so in the limit it leaves no trace and ordinary averaging describes everything. What produces a jump is not a large forcing imposed by hand but an *asymmetric potential* together with a scaling that keeps the oscillation slow while the asymmetry sharpens. In the model of [Section 2](#) the air supplies a wall that steepens without bound below the equilibrium while gravity is uniform above it, so a large oscillation rebounds in a short time and coasts for a long one; and because ε enters only as the *width of the band* in which the air is felt, a rebound occupies a vanishing fraction of its orbit while delivering an impulse of order one. [Lemmas 3.4](#) and [3.5](#) are those two statements. Concentration is the asymmetry of the well read at the right scale; it is not an assumption about the size of a force.

1.1 What this document does not prove

[Theorem 4.2](#) is Theorem C of the hierarchy, for one model. Three further statements belong to the programme and are not proved here; we name each with the role it plays, so that what is missing is visible rather than implied.

Theorem A: the decomposition itself. That every sequence of trajectories with $\varepsilon_n \rightarrow 0$ splits, along a subsequence, into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder. [Theorem 4.2](#) compares a trajectory with a trajectory of the limit system and does not assert this; what the Lyapunov functional of [Assumption 2.3](#) supplies is the bound on the number of profiles that the classical argument obtains by contradiction — it is [Eq. \(3.12\)](#), and it is already proved — so the decomposition is expected to follow quantitatively. Proving it is [objective 1](#).

The limit of the family in the weak topology. The dynamics become a linear constraint when the family is embedded in a space of occupation measures, and the solution set is then the support of a terminal measure. [Section 6](#) gives the definitions and the topology and states what the companion to [Theorem 4.2](#) would assert at that level; proving it is [objective 4](#).

The passage from the weak limit back to the support. A solution set is a support and not an integral. Granting the limit of the measures and the profiles it carries, that the limiting solution set is contained in the one those profiles generate is a further implication, and the hardest link in the chain; the proof below obtains the inclusion from trajectories directly and so does not cross it. Which route the general case should take is open, and it is the question we would most value an advisor's judgment on.

One further obligation is of a different kind and is deferred rather than open. A bound in closed form still has to be evaluated, and evaluation in floating point is where a guarantee is ordinarily lost; the setting in which it is not is validated interval arithmetic, in which every operation is rounded outward so that the computed interval encloses the exact result [3, 4, Ch. 3]. That is a question about evaluating a bound rather than about proving one, and it returns as [objective 5](#).

1.2 Terminology

Two fields meet here and several words mean different things in each. We fix them once.

term	what it means here
<i>orbit</i>	One closed curve of Eqs. (2.1) to (2.2) in the (z, v) plane: a single complete oscillation, in the sense of Lemma 2.7. <i>Not</i> a trajectory about the planet — the motions considered here are sub-orbital and decelerating throughout. A skip is one orbit; a multi-pass entry is a succession of them, with the amplitude decaying in between.
<i>pass, dip</i>	The part of an orbit below the section $z = 0$.
<i>profile</i> m_θ , <i>density</i> m_θ^ε	The <i>profile</i> is the ε -free function of Assumption 2.1, read in units of the band's own thickness; the <i>density</i> is the profile as the motion meets it, $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$. Hypotheses are placed on the profile, which is why they are uniform in ε ; the equations of motion carry the density.
<i>bubble</i>	A pass rescaled to its own width: a concentrating profile, whose net effect on the slow coordinates is a jump.
<i>skeleton</i>	What survives the limit: the arcs between passes, and the glide.
<i>residual</i>	The ball by which the inclusion is thickened.
<i>amplitude</i> W	The energy $\frac{1}{2}v^2 + \Phi_{c,\theta}(z)$ of the oscillation, and a Lyapunov functional for it. Not an amplitude in the sense of a maximum displacement.
<i>Lyapunov functional</i>	Two different objects wear this name and they do opposite jobs. The <i>dissipation</i> functional Θ_d of Assumption 2.3 and the amplitude W are non-increasing along the motion and measure what it spends; they are what the counts of Section 5 rest on. The <i>dual pair</i> (Φ, Ψ) of Section 6 is non-decreasing along the motion, and its superlevel set $\{\Psi \geq 1\}$ encloses the reachable set; that pair is the certificate, and exhibiting one is a proof whether or not it is ever evaluated numerically. The first kind is in hand; constructing the second is objective 4.
<i>averaging</i>	Always over one orbit, with a single phase (Lemma 4.5). Averaging over <i>several</i> phases, whose frequencies may cross, is a different problem, is not attempted anywhere below, and is the subject of objective 3.
<i>the oscillator</i>	Always the motion of Eqs. (2.1) to (2.2) — the altitude about the glide. The <i>attitude</i> oscillator of Section 7, three orders of magnitude faster, is named in full wherever it appears.

2 The model and its hypotheses

Fix an integer $d \geq 1$ and a horizon $T > 0$. The state is $x = (z, v, s)$, with $z \in \mathbb{R}$ an altitude, $v \in \mathbb{R}$ its rate, and $s \in \mathbb{R}^d$ a vector of slow coordinates. The system is

$$\dot{z} = v, \quad (2.1)$$

$$\dot{v} = c m_\theta^\varepsilon(z) - 1 - \kappa(s, c, \theta) v, \quad (2.2)$$

$$\dot{s} = F(s, c, \theta) + m_\theta^\varepsilon(z) \Gamma(s, c, \theta). \quad (2.3)$$

Here $c(\cdot)$ is the control and $\theta(\cdot)$ is a selection — a choice of admissible field from a bounded family. The small parameter $\varepsilon \in (0, 1]$ appears in one place only, inside the density: the *profile* m_θ is read on the layer's own scale,

$$m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon), \quad M_\theta^\varepsilon(z) = \varepsilon M_\theta(z/\varepsilon), \quad (2.4)$$

where M_θ is the antiderivative of m_θ with $M_\theta(0) = 0$, so that $(M_\theta^\varepsilon)' = m_\theta^\varepsilon$ and $M_\theta^\varepsilon(0) = 0$. The *potential* is

$$\Phi_{c,\theta}(z) = z - c M_\theta^\varepsilon(z), \quad \text{so} \quad \Phi'_{c,\theta}(z) = 1 - c m_\theta^\varepsilon(z), \quad (2.5)$$

and Eqs. (2.1) to (2.2) are a unit-mass particle in that potential with a weak damping. We write $\Phi_{c,\theta}$ without decoration, though it depends on ε : all of that dependence is the density's, and Lemma 3.4 is where it is unpacked. The *amplitude* is its energy,

$$W = \frac{1}{2}v^2 + \Phi_{c,\theta}(z). \quad (2.6)$$

The whole of Section 3 is an account of one object: the shape of this potential.

That ε sits inside the density and nowhere else is the whole of the scaling. The air is felt over a band of altitudes of thickness ε while the motion of interest is not thin: a large oscillation rises to an altitude of

order one and returns, so a pass occupies a vanishing *fraction* of each oscillation without the oscillation itself becoming fast. That is the configuration in which a pass concentrates, and [Lemmas 3.4](#) and [3.5](#) are the two statements that establish it — each of which also says what the other choice of scaling would have given instead.

Assumption 2.1 (H1: the wall). For each $\theta \in \Theta$ the profile $m_\theta \in C^1(\mathbb{R})$ satisfies $m_\theta > 0$ and $m'_\theta < 0$; $m_\theta(\xi) \rightarrow 0$ as $\xi \rightarrow +\infty$ with $\int_0^\infty m_\theta < \infty$; $m_\theta(\xi) \rightarrow \infty$ as $\xi \rightarrow -\infty$; and $1/c$ lies in the range of m_θ for every $c \in [c_{\min}, c_{\max}]$. We write $m_+ = \sup_\Theta m_\theta(0)$ and $M_\infty = \sup_\Theta \int_0^\infty m_\theta$, both finite and independent of ε .

Because m_θ decreases, m_+ bounds m_θ^ε at and above the section $z = 0$ for every ε , which is the only place a pointwise bound on the density is used below. Inside the band m_θ^ε is unbounded as $\varepsilon \rightarrow 0$ and no pointwise bound is taken: what is used there is its *impulse*, which [Assumption 2.3](#) bounds.

The two one-sided conditions are the whole physical content: the air thins upward without bound on the integral, and thickens downward without limit. Nothing in what follows uses the shape of m_θ beyond [Assumption 2.1](#), which is why θ may select it: an enclosure of an uncertain density is exactly a bounded family of admissible profiles.

Assumption 2.2 (H2: data). $[c_{\min}, c_{\max}] \subset (0, \infty)$, Θ and $S \subset \mathbb{R}^d$ are compact. The maps $F, \Gamma: S \times [c_{\min}, c_{\max}] \times \Theta \rightarrow \mathbb{R}^d$ and κ are continuous, bounded by $F_{\max}$, $\Gamma_{\max}$ and $\kappa_{\max}$, and Lipschitz in s uniformly in (c, θ) with constants κ_F, κ_Γ . Moreover $\kappa \geq \kappa_{\min} > 0$.

Assumption 2.3 (H3: a Lyapunov functional). There is $\Theta_d \in C^1(\mathbb{R}^d)$ and a constant $c_0 > 0$ with

$$\nabla \Theta_d(s) \cdot F(s, c, \theta) \leq 0, \quad \nabla \Theta_d(s) \cdot \Gamma(s, c, \theta) \leq -c_0,$$

for all (s, c, θ) . Write $M = (\max_S \Theta_d - \min_S \Theta_d)/c_0$.

A Lyapunov functional for the slow motion, non-increasing along every trajectory for every control and every selection. The bound on its total variation over the compact corridor is what converts monotonicity into a count: by [Eq. \(5.1\)](#) it bounds $\int m_\theta^\varepsilon d\tau$ uniformly in ε , and every count in [Section 5](#) descends from that one inequality. In the application it is the specific mechanical energy, which falls at the drag power because neither the lift nor the Coriolis force does work. [Section 7](#) notes a second monotone quantity the application supplies, carrying a different power of the density, and what it would cost to use.

Assumption 2.4 (H4: control). $c(\cdot)$ is absolutely continuous with values in $[c_{\min}, c_{\max}]$ and $|\dot{c}| \leq \dot{c}_{\max}$. (The sharper form, in which c is piecewise constant on intervals of length at least Δ , is [Remark 3.2](#).)

Assumption 2.5 (H5: corridor). Every trajectory remains on $[0, T]$ in a compact corridor $\mathcal{X} = \{|z| \leq z_{\max}, |v| \leq v_{\max}, s \in S\}$, and $x(0) \in \mathcal{X}_0 \subseteq \mathcal{X}$. Write $W_{\max} = \frac{1}{2}v_{\max}^2 + z_{\max}$ for the largest amplitude it admits.

The corridor is a fixed box, independent of ε , and it is the scaling of [Eq. \(2.4\)](#) that permits this: the oscillations of interest have amplitude of order one, so bounding them costs nothing that shrinks. The one quantity that needs care is the antiderivative, since M_θ^ε is unbounded below as $\varepsilon \rightarrow 0$ at fixed $z < 0$. It is not unbounded where a trajectory can reach. Wherever $\Phi_{c,\theta}(z) \leq W_{\max}$ we have $-c M_\theta^\varepsilon(z) = \Phi_{c,\theta}(z) - z \leq W_{\max} + z_{\max}$, and above the section $|M_\theta^\varepsilon| \leq \varepsilon M_\infty$, so

$$M_* := M_\infty + \frac{W_{\max} + z_{\max}}{c_{\min}} \tag{2.7}$$

bounds $|M_\theta^\varepsilon|$ on the reachable part of the corridor, uniformly in ε .

Remark 2.6 (What is uniform, and what the hypotheses are not). Every constant named above is a bound over $S \times [c_{\min}, c_{\max}] \times \Theta$. No constant below depends on the particular control, the particular selection, or the particular initial point: that is the sense in which the result is uniform over the admissible family, and it is why an enclosure of an uncertain field is the natural setting rather than a complication. Note also what is absent. There is no transversality condition, no non-grazing condition, and no assumption that a pass is a single dip — the last is [Lemma 2.7](#), a consequence of $m'_\theta < 0$. Every hypothesis is a condition on the data, checkable before any trajectory is computed.

Lemma 2.7 (One well, and one dip per period). *Under [Assumption 2.1](#) the potential $\Phi_{c,\theta}$ is strictly convex, with*

$$\Phi''_{c,\theta}(z) = -\frac{c}{\varepsilon} m'_\theta(z/\varepsilon) > 0, \quad (2.8)$$

and attains its minimum at the unique $z_(c, \theta) = \varepsilon \xi_*$, where ξ_* is the unique point with $c m_\theta(\xi_*) = 1$: the glide sits inside the band, at height of order ε . Consequently every sublevel set $\{\Phi_{c,\theta} \leq W\}$ is a bounded interval $[z_-(W), z_+(W)]$, and at frozen (c, θ) every bounded motion of [Eqs. \(2.1\) to \(2.2\)](#) with $\kappa = 0$ is a single closed orbit — one closed curve in the (z, v) plane, traversed periodically — on which z attains exactly one minimum and one maximum.*

Proof. Differentiating [Eq. \(2.5\)](#) twice and using [Eq. \(2.4\)](#) gives [Eq. \(2.8\)](#), which is positive because $c > 0$, $\varepsilon > 0$ and $m'_\theta < 0$. A strictly convex C^2 function has at most one stationary point; $\Phi'_{c,\theta} = 1 - c m'_\theta$ changes sign exactly once because m_θ is strictly decreasing from $+\infty$ to 0 and $1/c$ is in its range, so z_* exists and is unique. Strict convexity makes each sublevel set an interval, bounded because $\Phi_{c,\theta} \rightarrow +\infty$ at both ends. At frozen (c, θ) and $\kappa = 0$ the quantity [Eq. \(2.6\)](#) is constant, so the orbit is the level curve $\{\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)\}$, which is a single closed curve meeting $v = 0$ exactly at the two endpoints of the interval. Those are the one minimum and the one maximum of z . $\square$

[Lemma 2.7](#) is worth noting for what it is not: a hypothesis. That a pass is a single dip rather than a train of them is a property of the motion, and here it costs one sign on the density rather than a condition imposed on solutions.

3 Three regimes of one oscillator

Glide, phugoid and skip are not three systems but three energies in the single well of [Lemma 2.7](#). This section says which is which, and the statement that organises everything is that ε sets no shape — only a clock.

Lemma 3.1 (The amplitude law). *Along every trajectory of [Eqs. \(2.1\) to \(2.3\)](#),*

$$\dot{W} = -\kappa(s, c, \theta) v^2 - M_\theta^\varepsilon(z) \dot{c}, \quad (3.1)$$

whatever the profile m_θ and whatever ε . Consequently, with M_ the uniform bound [Eq. \(2.7\)](#),*

$$W(\tau) \leq W(0) + M_* \dot{c}_{\max} \tau \quad \text{for } \tau \in [0, T]. \quad (3.2)$$

Proof. Differentiate [Eq. \(2.6\)](#) along [Eqs. \(2.1\) to \(2.2\)](#):

$$\dot{W} = v\dot{v} + \Phi'_{c,\theta}(z) \dot{z} + \frac{\partial \Phi_{c,\theta}}{\partial c} \dot{c} = v(c m_\theta^\varepsilon - 1) - \kappa v^2 + (1 - c m_\theta^\varepsilon) v - M_\theta^\varepsilon(z) \dot{c},$$

where $\partial \Phi_{c,\theta} / \partial c = -M_\theta^\varepsilon(z)$ by [Eq. \(2.5\)](#). The two terms carrying $c m_\theta^\varepsilon$ cancel identically, which is [Eq. \(3.1\)](#); the cancellation uses only that the same $c m_\theta^\varepsilon$ appears in [Eq. \(2.2\)](#) and in $\Phi'_{c,\theta}$. Since $\kappa > 0$, [Eq. \(3.1\)](#) gives $\dot{W} \leq M_* \dot{c}_{\max}$, and [Eq. \(3.2\)](#) follows by integration. $\square$

So W is a Lyapunov functional for the fast motion, decaying except where the control pumps it, and [Assumption 2.4](#) is what makes [Eq. \(3.2\)](#) an *a priori* bound rather than a hypothesis about solutions. Together with the functional of [Assumption 2.3](#), one for the fast motion and one for the slow, these are the two monotone quantities on which every count below rests.

Remark 3.2 (The sharper form, from a finite guidance interval). If c is piecewise constant on intervals of length at least Δ — which is what a finite guidance interval supplies — then $\dot{c} = 0$ between switches and [Eq. \(3.1\)](#) reads $\dot{W} = -\kappa v^2$ exactly: W is non-increasing on each interval, and changes only by the jumps $|\Delta W| \leq M_* |\Delta c|$ at the at most T/Δ switching times. [Equation \(3.2\)](#) then holds with $M_* \dot{c}_{\max} \tau$ replaced by $M_* |\Delta c|_{\max} \lceil \tau / \Delta \rceil$, and LaSalle's argument applies interval by interval.

3.1 The orbit averages, and the time the oscillation may occupy

Recall from Section 1.2 that an *orbit* here is one closed curve in the (z, v) plane, never a trajectory about the planet. Two exact identities do most of the quantitative work, and both come from periodicity rather than from any property of the profile.

Lemma 3.3 (The mean density of an orbit). *Let $\langle \cdot \rangle$ denote the time average over one closed orbit of Eqs. (2.1) to (2.2) at frozen (c, θ) and $\kappa = 0$. Then*

$$\langle m_\theta^\varepsilon \rangle = \frac{1}{c}, \quad (3.3)$$

for every amplitude W , every profile satisfying Assumption 2.1 and every ε . Consequently the impulse of an orbit equals its period divided by c , and along any trajectory of Eqs. (2.1) to (2.3) the total time spent in closed oscillation satisfies

$$\sum_{\text{orbits}} (\text{period}) \leq c_{\max} M. \quad (3.4)$$

Proof. Integrate Eq. (2.2) over one period at $\kappa = 0$. The left side is $[v] = 0$ because v is periodic, so $\int (c m_\theta^\varepsilon - 1) d\tau = 0$, that is $c \int m_\theta^\varepsilon d\tau = \int d\tau$, which is Eq. (3.3) on dividing by the period. The impulse of the orbit is $\int m_\theta^\varepsilon d\tau = \langle m_\theta^\varepsilon \rangle \times (\text{period}) = (\text{period})/c$. Summing over orbits and using $\int_0^T m_\theta^\varepsilon d\tau \leq M$ from Assumption 2.3 gives Eq. (3.4). $\square$

Equation (3.3) deserves a word, because it is the reason the skeleton is the glide. Whatever the oscillation does, its time-averaged density equals the density at the equilibrium: the glide is not merely the bottom of the well but the *mean* of every oscillation about it. One consequence is used below: since $\langle m_\theta^\varepsilon \rangle$ does not depend on W , the averaged motion of the slow state does not either, so the amplitude is an output of the reduced description and not a driver of it.

3.2 What ε does, and what it does not

Lemma 3.4 (ε is a clock, and the amplitude it is read against). *Let $\xi = z/\varepsilon$ and $u = v/\sqrt{\varepsilon}$, and let $\Phi_{c,\theta}^0(\xi) = \xi - c M_\theta(\xi)$ be the ε -free potential of the profile. At frozen (c, θ) and $\kappa = 0$ the orbit through amplitude W is, in the (ξ, u) plane, the level set $\{\frac{1}{2}u^2 = \Omega - \Phi_{c,\theta}^0(\xi)\}$ at*

$$\Omega = W/\varepsilon, \quad (3.5)$$

which does not involve ε ; and $d\tau = \sqrt{\varepsilon} d\xi/u$, so every time measured along the orbit is $\sqrt{\varepsilon}$ times a quantity depending on (Ω, c, θ) alone. In particular the period and the time below the section $z = 0$ are

$$P = \sqrt{\varepsilon} P_0(\Omega, c, \theta), \quad P_{\text{dip}} = \sqrt{\varepsilon} D_0(\Omega, c, \theta). \quad (3.6)$$

Proof. Substituting $z = \varepsilon \xi$ in Eq. (2.4) gives $m_\theta^\varepsilon(z) = m_\theta(\xi)$ and $M_\theta^\varepsilon(z) = \varepsilon M_\theta(\xi)$, so $\Phi_{c,\theta}(z) = \varepsilon \Phi_{c,\theta}^0(\xi)$. Then $W - \Phi_{c,\theta}(z) = \varepsilon(\Omega - \Phi_{c,\theta}^0(\xi))$ and $\frac{1}{2}v^2 = \varepsilon \cdot \frac{1}{2}u^2$, which is the stated level set. Equation (2.1) gives $d\tau = dz/v = \varepsilon d\xi/(\sqrt{\varepsilon} u) = \sqrt{\varepsilon} d\xi/u$, so every elapsed time is $\sqrt{\varepsilon}$ times an integral over ξ of a function of (Ω, c, θ) . Taking it over the whole orbit and over its part with $\xi \leq 0$ gives Eq. (3.6). $\square$

So ε still sets no shape, only a clock — but the amplitude at which the clock is read is $\Omega = W/\varepsilon$, and that is what the scaling of Eq. (2.4) achieves. An oscillation of amplitude of order one is read at $\Omega \rightarrow \infty$, where the profile potential is dominated by the linear term and the orbit is a long ballistic arc with a brief rebound: P_0 grows like $2\sqrt{2\Omega}$ while D_0 grows only logarithmically, so by Eq. (3.6)

$$P \rightarrow 2\sqrt{2W}, \quad \frac{P_{\text{dip}}}{P} = \frac{D_0(W/\varepsilon)}{P_0(W/\varepsilon)} \rightarrow 0, \quad (3.7)$$

the period tending to a limit of order one and the dip fraction to zero. For the exponential profile $m_\theta = e^{-\xi}$ both are explicit, $D_0/P_0 = \log(4\Omega/c)/(2\Omega)(1 + o(1))$, so that

$$\frac{P_{\text{dip}}}{P} = \frac{\varepsilon}{2W} \log \frac{4W}{c\varepsilon} (1 + o(1)). \quad (3.8)$$

Had z been measured in layer thicknesses, Ω would have been W itself, D_0/P_0 would have been a fixed number at fixed amplitude, and nothing would have concentrated. This is the one place the choice of scaling does work, and [Lemma 3.5](#) is the other.

Lemma 3.5 (The impulse of a pass is its entry speed). *Let a dip occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$, and write $I = \int_{\alpha}^{\beta} m_{\theta}^{\varepsilon}(z) d\tau$ for its impulse and $P_{\text{dip}} = \beta - \alpha$. Then, exactly,*

$$cI = (v(\beta) - v(\alpha)) + P_{\text{dip}} + \int_{\alpha}^{\beta} \kappa v d\tau. \quad (3.9)$$

At frozen (c, θ) the dip is elastic — $\Phi_{c, \theta}(0) = 0$, so [Eq. \(2.6\)](#) gives $\frac{1}{2}v^2 = W$ at both crossings and $v(\beta) = |v(\alpha)| = \sqrt{2W}$ — and then

$$\left| I - \frac{2\sqrt{2W}}{c} \right| \leq \frac{(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}}{c_{\min}}. \quad (3.10)$$

Proof. Integrate [Eq. \(2.2\)](#) over $[\alpha, \beta]$ and rearrange: that is [Eq. \(3.9\)](#), with no approximation of any kind. For [Eq. \(3.10\)](#), substitute $v(\beta) - v(\alpha) = 2\sqrt{2W}$ and bound the last two terms of [Eq. \(3.9\)](#) by $(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}$, then divide by $c \geq c_{\min}$. $\square$

[Equation \(3.10\)](#) is the statement that makes a pass a profile rather than a perturbation. By [Eq. \(3.7\)](#) the dip occupies a vanishing fraction of its orbit; by [Eq. \(3.10\)](#) it nonetheless delivers an impulse of order one, fixed by the speed at which the oscillation entered. The two together are concentration, stated with constants: the duration goes to zero and the effect does not. In the alternative scaling the same calculation returns $I = O(\sqrt{\varepsilon})$, and a pass leaves no trace.

3.3 One threshold, and the two errors it sets

Everything the proof needs is now fixed by a single number. Fix $W_0 \in (0, W_{\max}]$ and call an orbit of amplitude W

- a *bubble* if $W \geq W_0$: its dip is short against its arc, so the pass is an event;
- *fast* if $W < W_0$: its period is short against the slow drift, so it may be averaged.

The two classes exhaust the orbits by construction — there is no intermediate band and no hypothesis that one be empty — and the price is that the quality of each approximation is now something to be computed rather than assumed. Let

$$\vartheta_1 = \sup \left\{ \frac{D_0}{P_0}(\Omega, c, \theta) : \Omega \geq W_0/\varepsilon \right\}, \quad \vartheta_2 = \sqrt{\varepsilon} \sup \{ P_0(\Omega, c, \theta) : \Omega \leq W_0/\varepsilon \}, \quad (3.11)$$

the suprema over $c \in [c_{\min}, c_{\max}]$ and $\theta \in \Theta$ as well: ϑ_1 is the largest dip fraction any bubble can have, and ϑ_2 the longest period any fast orbit can have.

Lemma 3.6 (Both thresholds are small, and how small). *Under [Assumptions 2.1](#) and [2.5](#) the suprema in [Eq. \(3.11\)](#) are finite, and as $\varepsilon \rightarrow 0$ at fixed W_0 ,*

$$\vartheta_1 \rightarrow 0, \quad \vartheta_2 \rightarrow 2\sqrt{2W_0}.$$

For the exponential profile the supremum defining ϑ_1 is attained at the left endpoint once ε is small, and

$$\vartheta_1 = \frac{\varepsilon}{2W_0} \log \frac{4W_0}{c\varepsilon} (1 + o(1)).$$

Proof. Finiteness is [Eq. \(3.6\)](#) with Ω confined by [Eq. \(3.5\)](#) and [Assumption 2.5](#). For ϑ_1 , the supremum is over $\Omega \geq W_0/\varepsilon \rightarrow \infty$, and $D_0/P_0 \rightarrow 0$ there by [Eq. \(3.7\)](#); for the exponential profile D_0/P_0 is eventually decreasing in Ω by [Eq. \(3.8\)](#), so the supremum is the value at $\Omega = W_0/\varepsilon$, which is [Eq. \(3.8\)](#) at $W = W_0$. For ϑ_2 , write $\Omega_c = W_0/\varepsilon$. Since $P_0(\Omega) \rightarrow \infty$ with Ω , and P_0 is continuous, the supremum over $\Omega \leq \Omega_c$ is attained at Ω_c once Ω_c is large enough that $P_0(\Omega_c)$ exceeds the maximum of P_0 on any fixed bounded range — which happens for all small ε . Then $\vartheta_2 = \sqrt{\varepsilon} P_0(\Omega_c) \rightarrow 2\sqrt{2W_0}$ by [Eq. \(3.7\)](#). No monotonicity of P_0 or of D_0/P_0 in Ω is used anywhere, which is why both thresholds are defined as suprema. $\square$

It is worth being explicit that nothing here is assumed. A construction that classified orbits by their dip fraction and their period separately would need the two classes to cover every orbit, and that is a condition relating the thresholds to ε — the one condition of such a construction that is not a condition on the data. Classifying by amplitude alone removes it: the classes cover the orbits for every W_0 and every ε , and what would have been a hypothesis to check is instead the two quantities Eq. (3.11) to evaluate.

Remark 3.7 (What the threshold trades). The two errors move oppositely in W_0 , and choosing between them is Remark 4.3. Raising W_0 admits fewer bubbles, each with a shorter dip relative to its arc, so ϑ_1 falls; but it leaves more orbits to be averaged over a longer period, so ϑ_2 rises like $2\sqrt{2W_0}$. Lowering it does the reverse. Since ϑ_1 carries a factor $\varepsilon \log(1/\varepsilon)$ and ϑ_2 does not, the best threshold shrinks with ε . The count of bubbles moves with it: by Eq. (3.10) each has impulse at least $2\sqrt{2W_0}/c_{\max}$, so Assumption 2.3 admits at most

$$N_* = \frac{c_{\max} M}{2\sqrt{2W_0}} \quad (3.12)$$

of them — a bound free of ε at fixed W_0 , which is the profile count the classical argument obtains only by contradiction, and which grows as W_0 is lowered to buy a smaller residual.

3.4 The bubble

Definition 3.8 (Pass map). For a bubble, let (v^-, s^-) be its data where it last crosses $z = 0$ downward and (v^+, s^+) where it next crosses upward, computed from Eqs. (2.1) to (2.2) together with the *inner* slow equation $\dot{s} = m_\theta^\varepsilon(z) \Gamma(s, c, \theta)$ — that is, Eq. (2.3) with the drift F dropped, whose contribution over a dip is at most $F_{\max} P_{\text{dip}}$. The *pass map* is

$$J(v^-, s^-) = \{ (v^+, s^+) : \text{measurable } c(\cdot), \theta(\cdot) \text{ on the dip} \}.$$

Lemma 3.9 (The bubble is nearly elastic, and bounded in closed form). *Let $v^- \leq -v_0 < 0$ with $v_0 = \sqrt{2W_0}$, which every bubble satisfies because $\frac{1}{2}(v^-)^2 = W \geq W_0$. Along every orbit of Definition 3.8:*

- (a) *at frozen (c, θ) the dip is elastic: since $\Phi_{c, \theta}(0) = 0$, Eq. (2.6) gives $\frac{1}{2}(v^\pm)^2 = W$ at both crossings, so $v^+ = |v^-|$ exactly;*
- (b) *in general W drifts across the dip by at most $(\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}$ by Eq. (3.1), so*

$$|v^+ - |v^-|| \leq \frac{(\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}}{v_0};$$

- (c) *the dip reaches no lower than the z_- with $\Phi_{c, \theta}(z_-) = W$, which exists and is unique by Lemma 2.7, and $|z_-| \rightarrow 0$ with ε ;*
- (d) *its impulse I obeys Eq. (3.10), and $|s^+ - s^-| \leq \Gamma_{\max} I$.*

Proof. (a) On the dip the orbit is the level set of Eq. (2.6), so $\frac{1}{2}v^2 = W - \Phi_{c, \theta}(z)$ throughout; at $z = 0$ the potential vanishes because $M_\theta^\varepsilon(0) = 0$, giving $\frac{1}{2}(v^\pm)^2 = W$ with the same W at both crossings. (b) Away from frozen data W is no longer constant, and Eq. (3.1) bounds its rate by $\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}$ with M_* from Eq. (2.7); integrating over the dip and dividing by $|d(\frac{1}{2}v^2)/dv| = |v| \geq v_0$ gives the displayed bound. (c) is Lemma 2.7, and $z_- = \varepsilon \xi_-(\Omega)$ with $|\xi_-| = O(\log \Omega)$ by Lemma 3.4. (d) is Lemma 3.5 together with $s^+ - s^- = \int_{\text{dip}} m_\theta^\varepsilon \Gamma \, d\tau$. $\square$

Lemma 3.10 (The pass map, and how an error crosses it). *Let two entry data for the same (c, θ) differ by $(\delta v, \delta s)$, with both $|v^-| \geq v_0$. Then the exits differ by $(\delta v^+, \delta s^+)$ with*

$$|\delta v^+| = |\delta v|, \quad |\delta s^+| \leq e^{\kappa_\Gamma I} |\delta s| + \frac{2\Gamma_{\max}}{c_{\min}} |\delta v|, \quad (3.13)$$

where I is the impulse of the dip. Consequently J is Lipschitz in the Hausdorff distance, and — what the proof needs — the factor multiplying the slow error is exponential in that dip's own impulse.

Proof. Parametrise the dip by the delivered impulse $\mu = \int m_\theta^\varepsilon d\tau \in [0, I]$, in which $ds/d\mu = \Gamma(s, c, \theta)$, Lipschitz in s with constant κ_Γ and involving neither z nor v ; Gronwall over the horizon I gives the factor $e^{\kappa_\Gamma I}$ on δs . The speed is an isometry: by Lemma 3.9(a) $v^+ = |v^-|$ at frozen data, so $|\delta v^+| = |\delta v^-|$ with no amplification — there is no Gronwall to do on v , and in particular no factor $\varepsilon^{-1/2}$, which the inner equation for v carries but the exit condition removes. What remains is the dependence of the exit on δv through the horizon itself: by Eq. (3.10), $|\partial I/\partial v^-| \leq 2/c_{\min}$ up to $O(P_{\text{dip}})$, and $|\partial s^+/\partial I| \leq \Gamma_{\max}$, which is the second term of Eq. (3.13). Taking suprema over (c, θ) gives the Hausdorff statement. $\square$

The form of Eq. (3.13) is what matters downstream and is worth isolating. The slow error is amplified by $e^{\kappa_\Gamma I}$, a factor that tends to 1 as the dip's impulse tends to zero, so the product of these factors over any collection of bubbles is $e^{\kappa_\Gamma \Sigma I}$ and Assumption 2.3 bounds the exponent whatever the number of bubbles. The coupling from δv enters *additively* and not as a factor, and the speed error is not amplified at all; so the two cannot compound into a product over the passes. A bound stated as a single Lipschitz constant for J would lose exactly this, since such a constant exceeds 1 by a fixed amount and its power would diverge with the number of passes.

4 The limit system, and the theorem

Definition 4.1 (The limit system). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. The limit system $\mathcal{H}$ has state (z, v, s) on the arcs and (s) on the glide, with three transitions.

Arc (flow). $\dot{z} = v$, $\dot{v} = -1 - \kappa v$, $\dot{s} = F(s, c, \theta)$, while $z \geq 0$: the ballistic arc, on which the air is not felt.

Bubble (jump). At $z = 0$ with $v \leq -v_0$ the state is replaced by $(0, v^+, s^+)$ with $(v^+, s^+) \in J(v^-, s^-)$, consuming no time.

Glide (averaged flow). For $W < W_0$, $\dot{s} = F(s, c, \theta) + \frac{1}{c} \Gamma(s, c, \theta)$, by Eq. (3.3), and z is not carried.

$\mathcal{R}_{\mathcal{H}}(t)$ is the set of slow states reached at time t from $\mathcal{X}_0$ over all admissible c, θ and all selections from J , and $\mathcal{R}_{\mathcal{H}}(E) = \bigcup_{t \in E} \mathcal{R}_{\mathcal{H}}(t)$.

Note what the glide mode does *not* carry: by Eq. (3.3) the averaged slow field is independent of the amplitude, so no action coordinate is needed. The amplitude is an output of the reduced description. Note also that $\mathcal{H}$ is free of ε : the arcs are ballistic, the glide is averaged, and the only trace of the layer is the pass map, which Eq. (3.10) fixes to leading order in terms of the entry speed alone.

We measure the residual in $|(z, v, s)| = |z| + |v| + |s|$ and write $\mathbb{B}(\varrho)$ for the corresponding ball, $\oplus$ for the Minkowski sum. The constants are

$$C_f = F_{\max} + m_+ \Gamma_{\max}, \quad \kappa_0 = \kappa_F + \frac{\kappa_\Gamma}{c_{\min}}, \quad G = e^{\kappa_0 T + \kappa_\Gamma M}, \quad C_A = \frac{\Gamma_{\max} + \kappa_\Gamma C_f T}{c_{\min}}, \quad (4.1)$$

with $m_+ = \sup_\Theta m_\theta(0)$ from Assumption 2.1 and M from Assumption 2.3. Every one of them is free of ε . That is not an accident of bookkeeping but the one discipline the proof keeps: wherever the density appears inside the band, where it is unbounded as $\varepsilon \rightarrow 0$, it is bounded by its *impulse* and never pointwise, and Assumption 2.3 bounds every impulse that occurs. The pointwise bound m_+ is used only above the section, where Assumption 2.1 makes it uniform.

Theorem 4.2 (Outer inclusion with an explicit residual). Assume Assumptions 2.1 to 2.5 and fix $W_0 \in (0, W_{\max}]$. Let ϑ_1, ϑ_2 be the thresholds Eq. (3.11), N_* the count Eq. (3.12), and A_ε the total impulse felt on the arcs, for which $A_\varepsilon \leq \varepsilon M_\infty c_{\max} M / (2W_0)$. Then

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\mathcal{R}_{\mathcal{H}}([T - \varrho, T + \varrho])}_{\text{skeleton, bubbles and glide}} \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}}, \quad (4.2)$$

with

$$\varrho = G \left(\underbrace{F_{\max} c_{\max} M \vartheta_1}_{\text{the drift over the dips}} + \underbrace{C_A \vartheta_2}_{\text{averaging the fast orbits}} + \underbrace{\frac{2\Gamma_{\max}}{c_{\min}} N_* c_{\max} A_\varepsilon}_{\text{the air the arcs discard}} \right). \quad (4.3)$$

All constants are those of Assumptions 2.1 to 2.5 through Eq. (4.1); none depends on the control, the selection, the initial point or ε .

Three remarks on the statement. The inclusion is over a short interval of times, of half-width the residual, because a dip of positive duration is compared with an instantaneous jump. Each of the three terms of Eq. (4.3) is the error of exactly one approximation and is traced to one estimate below, so the theorem is modular: an improvement to any one of the three lemmas improves the residual without touching the others. And the theorem is stated at a *fixed* threshold, so ϱ is a number and not a rate; what the thresholds do as $\varepsilon \rightarrow 0$ is Remark 4.3.

Remark 4.3 (The threshold at its best, and the rate that follows). By Lemma 3.6 the first term of Eq. (4.3) falls like $\varepsilon \log(1/\varepsilon)/W_0$ and the second rises like $\sqrt{W_0}$, while the third falls like $\varepsilon/W_0^{3/2}$, so there is a best threshold and it shrinks with ε . Balancing the second against the third gives $W_0 \asymp \sqrt{\varepsilon}$ and

$$\varrho = O(\varepsilon^{1/4}), \quad (4.4)$$

at which threshold the first term is $O(\sqrt{\varepsilon} \log(1/\varepsilon))$ and is not binding — so the logarithm of Eq. (3.8) does not reach the rate. We record Eq. (4.4) for the order of magnitude and claim nothing about its sharpness: it is what balancing three crude bounds gives. Which of the three is the real obstruction is a quantitative question we would want settled early, and the third — the air felt on the arcs, which is an artefact of cutting the pass at $z = 0$ rather than at the edge of the band — is the one we would expect to improve first.

Remark 4.4 (At a fixed ε , and what the thresholds then cost). Nothing above requires ε to be small: Eq. (4.3) holds at every ε and every W_0 , and at a fixed physical ε it is evaluated rather than estimated. What it then says is that the residual is small only if ϑ_1 and ϑ_2 both are, and Section 7 shows that for entry they are not: no threshold makes both the dips short and the fast periods short. The remedy is not a new error term but a narrower description. The orbits near W_0 , which neither approximation serves, may be *propagated exactly* — carrying the fast pair (z, v) and the unapproximated field Eqs. (2.1) to (2.3) — and this adds nothing to ϱ because nothing is approximated. What it costs is dimension and amplification: the reduced problem carries $2 + d$ coordinates instead of d during those windows, and G acquires the factor $e^{\kappa_0 T_E}$ for their total duration T_E , which is at most $c_{\max} M$ by Eq. (3.4) *whatever the threshold*. So the threshold trades sharpness against dimension, not sharpness against sharpness.

Lemma 4.5 (Averaging over one orbit, with its constant). *Let s solve Eq. (2.3) along a fast orbit of period P , at frozen (c, θ) , and let $\bar{s}$ solve $\dot{\bar{s}} = F(\bar{s}, c, \theta) + \frac{1}{c}\Gamma(\bar{s}, c, \theta)$ with $\bar{s}(0) = s(0)$. Then on $[0, T]$*

$$|s - \bar{s}| \leq G C_A P$$

with G and C_A as in Eq. (4.1); the factor G is the amplification over the horizon, and it is the same G that Eq. (4.3) collects once in front of every term, so C_A itself carries none of it. The data are frozen because Eq. (3.3), on which the proof turns, holds at frozen (c, θ) ; over one fast orbit c moves by at most $\dot{c}_{\max} P$ by Assumption 2.4, so releasing them adds a term of the same order $O(P)$ and is absorbed by enlarging C_A by $\dot{c}_{\max} M_$.*

Proof. Put $e = s - \bar{s}$. Subtracting, $\dot{e} = [F(s) - F(\bar{s})] + m_\theta^\varepsilon[\Gamma(s) - \Gamma(\bar{s})] + (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})$, so $|\dot{e}| \leq (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon)|e| + |(m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})|$. Let $R(\tau) = \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})$. By Eq. (3.3) the integrand has zero mean over each period, so R is periodic with $R(0) = R(P) = 0$ and

$$|R| \leq \frac{1}{2} \int_0^P \left| m_\theta^\varepsilon - \frac{1}{c} \right| d\tau \leq \frac{1}{2} \left(\int_0^P m_\theta^\varepsilon d\tau + \frac{P}{c} \right) = \frac{P}{c},$$

the last step by Eq. (3.3) again. This is the one place the bound must not be taken pointwise: m_θ^ε is unbounded inside the band as $\varepsilon \rightarrow 0$, while its integral over a period is exactly P/c . Integrating the last term by parts,

$$\left| \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s}) \right| \leq |R| \Gamma_{\max} + |R| \int_0^\tau \left| \frac{d}{d\tau} \Gamma(\bar{s}) \right| \leq \frac{P}{c} (\Gamma_{\max} + \kappa_\Gamma C_f T),$$

using $|\dot{\bar{s}}| \leq C_f$. Gronwall's inequality in integrated form multiplies this by $\exp \int_0^T (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon) \leq e^{\kappa_F T + \kappa_\Gamma M} \leq G$, again by Assumption 2.3 and not by a pointwise bound, which gives the claim with the C_A of Eq. (4.1). Note that the error does not accumulate over the successive fast orbits: the oscillating term has zero mean by Eq. (3.3), so its primitive is bounded over the whole horizon and the bound is of the order of *one* period, not of their sum. $\square$

5 Proof of Theorem 4.2

Fix $\varepsilon \in (0, \varepsilon_1]$, an admissible trajectory $x_\varepsilon = (z, v, s)$ on $[0, T]$ with control $c(\cdot)$ and selection $\theta(\cdot)$, and $x_\varepsilon(0) \in \mathcal{X}_0$.

Step 1: the budget

By [Assumption 2.3](#), $\frac{d}{d\tau} \Theta_d(s) = \nabla \Theta_d \cdot F + m_\theta^\varepsilon \nabla \Theta_d \cdot \Gamma \leq -c_0 m_\theta^\varepsilon$, so

$$\int_0^T m_\theta^\varepsilon(z) d\tau \leq M, \quad (5.1)$$

for every control, every selection and every ε .

Step 2: the orbits, how long they last, and how many there are

By [Lemma 2.7](#) the frozen motion is a single closed orbit through each amplitude, and by [Lemma 3.1](#) the amplitude changes only through the damping and the control, with the *a priori* bound [Eq. \(3.2\)](#). So $[0, T]$ decomposes into arcs, on which the air is scarcely felt, and orbits. By [Lemma 3.3](#) each orbit's impulse is its period divided by c , so by [Eq. \(5.1\)](#) the total time in orbits is at most $c_{\max} M$ and the total impulse at most M .

Three counts follow, and all three come from [Eq. \(5.1\)](#) alone. First, the summed dip durations: a bubble has dip fraction at most ϑ_1 by [Eq. \(3.11\)](#), so $P_{\text{dip}} \leq \vartheta_1 P$, and summing over bubbles with $\sum P \leq c_{\max} M$ from [Eq. \(3.4\)](#),

$$\sum_{\text{bubbles}} P_{\text{dip}} \leq \vartheta_1 c_{\max} M. \quad (5.2)$$

Second, the number of bubbles: by [Eq. \(3.10\)](#) each has impulse at least $2\sqrt{2W_0}/c_{\max}$, so [Eq. \(5.1\)](#) admits at most the N_* of [Eq. \(3.12\)](#). Third, the impulse the arcs discard: on an arc $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$ with $z \geq 0$, so by [Lemma 3.4](#) $\int_{\text{arc}} m_\theta^\varepsilon d\tau = \sqrt{\varepsilon} \int_0^{\xi^+} m_\theta(\xi) d\xi/u \leq \varepsilon 2M_\infty/\sqrt{2W}$, and summing over at most N_* orbits,

$$A_\varepsilon := \sum_{\text{arcs}} \int m_\theta^\varepsilon d\tau \leq \frac{\varepsilon M_\infty c_{\max} M}{2W_0}. \quad (5.3)$$

[Equations \(5.2\)](#) and [\(5.3\)](#) are the two estimates that make the construction work, and they are worth reading twice. The number of bubbles grows as the threshold is lowered and each dip shortens like $\varepsilon \log(1/\varepsilon)$, and [Eq. \(5.2\)](#) says the product is controlled by ϑ_1 with no ε left in it; the arcs each feel an impulse of order ε , and [Eq. \(5.3\)](#) says the total is of order ε/W_0 . Neither is a compactness argument and neither could be obtained from one.

Step 3: every orbit is a bubble or fast

This is now a matter of definition and not a hypothesis: an orbit has $W \geq W_0$ or it does not, so the two treatments of [Definition 4.1](#) exhaust the motion for every threshold and every ε . What has to be quantified instead is how good each treatment is, and that is [Lemma 3.6](#): the worst dip fraction among the bubbles is ϑ_1 and the worst period among the fast orbits is ϑ_2 , both finite, computable from the profile and the data, and both small for small ε at fixed W_0 .

Step 4: a bubble is a jump, up to the drift over its dip

Let a bubble occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$. By [Definition 3.8](#) the exit data of the inner problem from $(v(\alpha), s(\alpha))$ differ from $(v(\beta), s(\beta))$ only through the one term dropped in [Eq. \(2.3\)](#), namely the drift F , whose integral over the dip is at most $F_{\max} P_{\text{dip}}$. Hence

$$(v(\beta), s(\beta)) \in J(v(\alpha), s(\alpha)) \oplus \mathbb{B}(F_{\max} P_{\text{dip}}), \quad (5.4)$$

and $z(\alpha) = z(\beta) = 0$ exactly, so the altitude contributes nothing. It is $F_{\max}$ and not C_f that appears here: the density term of Eq. (2.3) is retained by the inner problem, which is the only reason this step survives a density that is unbounded inside the band.

Step 5: a fast orbit is the averaged glide, up to $C_A P$

This is Lemma 4.5, with $P \leq \vartheta_2$ by Eq. (3.11), so the error is at most $G C_A \vartheta_2$ — the G being the horizon amplification that Eq. (4.3) collects in front of every term, counted here once and not again in Step 6. Note which averaging this is. The model has one fast phase, so the average is over a single closed orbit, there is no resonance condition to satisfy, and the constant is the one computed in Lemma 4.5. Averaging over *several* phases, whose frequencies may cross, is a different problem and is not attempted anywhere in this document; it is the subject of objective 3, where we propose not to attempt it at all.

Step 6: assembling, and why the amplification is bounded

Construct $x_{\mathcal{H}}$ by following $\mathcal{H}$ with the same (c, θ) , jumping at each bubble with the point of J nearest to x_{ε} 's exit and flowing with the averaged field on each fast orbit. Write δ for the discrepancy and track its two components separately, which is what Eq. (3.13) is for.

The slow error is amplified in two places and in both the factor is exponential in an impulse. On the arcs and the glide both motions obey fields Lipschitz with constant κ_0 , giving $e^{\kappa_0 T}$ over the horizon. Across a bubble, Eq. (3.13) gives the factor $e^{\kappa_{\Gamma} I}$ for a bubble of impulse I , so the product over all bubbles is

$$\prod_{\text{bubbles}} e^{\kappa_{\Gamma} I_k} = e^{\kappa_{\Gamma} \sum_k I_k} \leq e^{\kappa_{\Gamma} M}$$

by Eq. (5.1). Together the two are the G of Eq. (4.1). This is the second place the budget does work that no compactness argument could: the amplification compounds over every pass, however many there are, and is nonetheless bounded, because each factor is exponential in that pass's own impulse and the impulses are budget-limited. The step turns on the *form* of Eq. (3.13) and not merely on its size. Had the passage of an error across a bubble been recorded as a single Lipschitz constant for J , that constant would have exceeded 1 by a fixed amount and its product over the passes would have diverged as the threshold was lowered.

The speed error is not amplified at all, by Eq. (3.13): the bubble is an isometry in v . It is only *injected*, on the arcs, where x_{ε} feels the density that $x_{\mathcal{H}}$ does not, at a total cost of $c_{\max} A_{\varepsilon}$ by Eq. (5.3). It reaches the slow error through the additive coupling of Eq. (3.13), once per bubble, for a total of at most $(2\Gamma_{\max}/c_{\min}) N_{*} c_{\max} A_{\varepsilon}$ — the third term of Eq. (4.3), and the only one in which the number of passes appears.

Collecting Eqs. (5.2) and (5.4) for the bubbles, Step 5 for the fast orbits and Eq. (5.3) for the arcs,

$$|x_{\varepsilon}(T) - x_{\mathcal{H}}(\hat{T})| \leq G \left(F_{\max} \sum_{\text{bubbles}} P_{\text{dip}} + C_A \vartheta_2 + \frac{2\Gamma_{\max}}{c_{\min}} N_{*} c_{\max} A_{\varepsilon} \right),$$

which with Eq. (5.2) is the ϱ of Eq. (4.3). The arrival time $\hat{T}$ differs from T because x_{ε} spends the dips where $x_{\mathcal{H}}$ spends no time, and by Eq. (5.2) that discrepancy is at most $\vartheta_1 c_{\max} M$, which is at most $\varrho/(G F_{\max})$ and so at most ϱ whenever $G F_{\max} \geq 1$. Hence $\hat{T} \in [T - \varrho, T + \varrho]$ and Eq. (4.2) follows. $\square$

Remark 5.1 (Where each hypothesis was used). Assumption 2.1 was used twice: in Lemma 2.7, from which everything structural descends — one well, one dip per period, and hence countable passes — and for the uniform bound m_{+} on the density above the section. Assumption 2.3 was used in Step 1, and everything quantitative descends from it: the summed dips Eq. (5.2), the pass count Eq. (3.12), the arc impulse Eq. (5.3), the total time in orbits Eq. (3.4), the bounded amplification of Step 6, and the two places in Lemma 4.5 where a pointwise bound on the density would have failed. Assumption 2.4 was used once, for the *a priori* amplitude bound Eq. (3.2). Assumptions 2.2 and 2.5 supply the constants. There are five hypotheses and no sixth: that the two treatments cover every orbit is Lemma 3.6, not an assumption. No compactness argument and no measure appears anywhere above.

6 The weak formulation

[Theorem 4.2](#) is a statement about trajectories, and its proof uses no measures. The weak formulation is nevertheless where the question belongs, for two reasons: it is what makes the solution set of a nonconvex family a tractable object at all, by rendering the dynamics a linear constraint; and it is the setting in which the loss of compactness that motivates the whole decomposition is visible. We give the definitions, so that what is open can be stated precisely.

For a trajectory of [Eqs. \(2.1\) to \(2.3\)](#) with control $c(\cdot)$ and selection $\theta(\cdot)$, write $\mathcal{U} = [c_{\min}, c_{\max}]$ for the control set of [Assumption 2.2](#). The *occupation measure* is the element $\mu_\varepsilon \in \mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta)$ defined by

$$\int \varphi d\mu_\varepsilon = \int_0^T \varphi(t, x_\varepsilon(t), c(t), \theta(t)) dt \quad \text{for } \varphi \in C([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta),$$

together with the initial and terminal measures μ_0, μ_T carried by $x_\varepsilon(0)$ and $x_\varepsilon(T)$. The topology is the weak-* topology, that is, the topology of the dual of that space of continuous functions; the base is a compact metric space by [Assumption 2.2](#), so the topology is metrizable and bounded sets are relatively compact. In these terms the dynamics are a *linear* constraint: for every $\varphi \in C^1$,

$$\int \varphi(T, \cdot) d\mu_T - \int \varphi(0, \cdot) d\mu_0 = \int \left(\frac{\partial \varphi}{\partial t} + \nabla \varphi \cdot (v, c m_\theta^\varepsilon - 1 - \kappa v, F + m_\theta^\varepsilon \Gamma) \right) d\mu_\varepsilon, \quad (6.1)$$

which is the Liouville equation of the family, and the reachable set at time T is the union of the supports of the terminal measures that satisfy it. This is the occupation-measure programme of Lasserre, Henrion and collaborators [\[5, 6, 7\]](#), and an outer bound comes from its dual: a pair (Φ, Ψ) with Φ non-decreasing along the admissible motions and at least one on $\mathcal{X}_0$, and Ψ majorising Φ at time T , encloses the reachable set in $\{\Psi \geq 1\}$.

What [Theorem 4.2](#) does is to hand this machinery a different problem. After the theorem, the object to be enclosed is $\mathcal{R}_\mathcal{H}$: a problem in a few coordinates with a flow, one guard $z = 0$, a set-valued reset J , and a number of jumps bounded by the Lyapunov functional through [Eq. \(3.12\)](#). That is exactly the hybrid form for which the occupation-measure relaxation is known [\[8, 9\]](#): a measure on each guard carries the mass from one mode to the next, and the certificate Φ must be non-decreasing along the arc flow and across the reset. The singular limit has been discharged by [Section 5](#), which is elementary, and what is left on $\mathcal{H}$ is a problem in a few coordinates. Whether the limit is better taken in this topology, with the decomposition extracted from the measures, or at the level of trajectories as in [Section 5](#), is the open question of [Section 1.1](#), and the two routes are not known to be equivalent.

Remark 6.1 (What the reduction buys, and what now binds). The reduction is often put as a dimension count, and the count is real: the original problem carries thirteen coordinates and more, while $\mathcal{H}$ carries a few. But the dimension is not the binding constraint, because what is to be produced is a pair (Φ, Ψ) and not a mesh. A grid method for the Hamilton–Jacobi–Isaacs equation is exponential in the dimension and so has a hard ceiling near four or five states; a dual pair is *exhibited and verified*, and the dimension enters through the difficulty of finding one — and, if it is sought by a moment or sum-of-squares relaxation, through the size of the resulting programme — rather than as an exponential wall. Two consequences. The coordinate that [Section 7](#) would spend to use a second monotone quantity, and the fast pair that [Remark 4.4](#) carries through the orbits near the threshold, are costs rather than obstructions. And what does bind is the pass map: the condition on Φ at a jump is that it not decrease across it for *every* point of $J(v^-, s^-)$, so no certificate can be written down until J is enclosed. That is [objective 2](#), and it is not a question of dimension.

Two things are not done. [Theorem 4.2](#) compares trajectories and not measures, and the companion statement at the level of measures is [objective 4](#). And nothing above makes $\mathcal{R}_\mathcal{H}$ tight: an outer bound on it is an outer bound on $\mathcal{R}_\varepsilon(T)$ up to the residual, by [Eq. \(4.2\)](#), but how much is lost in passing from the one to the other is a separate question from how much is lost in the limit.

7 What hypersonic entry adds

The application is hypersonic entry: the reachable set of an entry vehicle, for any vehicle and any corridor, with the vehicle and its corridor entering only as data. Entry is not an arbitrary illustration, because it

supplies the singular perturbation, the limit system and the concentrating events in one system. Nor is it already covered: the exact method solves a Hamilton–Jacobi–Isaacs equation on a grid, capped in practice near four or five states [10, 11], while a rigid-body entry vehicle carries thirteen and more once ablation and the aerothermoelastic modes are coupled [12]; and set-propagation methods return rigorous enclosures at ten to twenty states [13, 14, 15], but of one instance — one vehicle, one initial box, one horizon — with no explicit dependence on the corridor or on the width of the aerodynamic uncertainty, which is what a statement uniform over a family requires.

7.1 The correspondence

Measure altitude in planetary radii above the equilibrium glide, so that the atmospheric density is $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$ with the profile m_θ decreasing and read in scale heights, and let

$$\varepsilon = H_s/R_e$$

be the thin-atmosphere ratio, whose square root is the ratio of the oscillation period about the glide to the slow time $\sqrt{R_e/g}$. Then Eqs. (2.1) to (2.2) are the vertical motion of the vehicle: v is the rate of climb in those units, c is the vertical lift fraction $(L/D) \cos \sigma$ normalised so that $c m_\theta^\varepsilon = 1$ is the glide balance — lift against weight less centrifugal relief — and κ is the phugoid damping, which does not scale with ε . The slow state s collects the energy, the range and the heading, F carries the gravitational terms and Γ the drag. The selection θ chooses the density shape and the aerodynamic coefficients from an enclosure, so that Eqs. (2.1) to (2.3) are a differential inclusion rather than a single field [16, 17], and Remark 2.6 is the statement that the bound holds over the whole enclosure. Here the vehicle enters only through dimensionless ratios — c , κ , the budget constants — each quantified over a range for a class, never evaluated at a design.

The identification of the three regimes is the content of the model rather than an analogy. The minimum of the well is the equilibrium glide; the small oscillation about it is the phugoid, of period $\sqrt{\varepsilon}$ on the slow time scale and damping ratio of the same order, so that it persists for the whole entry — and it lives inside the band, at height of order ε , by Lemma 2.7; and a large oscillation is a skip, of period of order one, rebounding from the exponential wall in a time of order $\varepsilon \log(1/\varepsilon)$ and coasting on a long ballistic arc. That the dip is short and the arc long is Lemma 3.4 together with the asymmetry of $\Phi_{c,\theta}$, and that the rebound nevertheless delivers an impulse of order one is Lemma 3.5; nothing more enters.

Assumption 2.3 is the hypothesis that transfers exactly. The specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, satisfies $\dot{\mathcal{E}} = -P$ with $P = DV/m \geq 0$ the drag power, because neither the lift nor the Coriolis force does work. Taking $\Theta_d = \mathcal{E}$ gives $\nabla \Theta_d \cdot F = 0$ exactly, and $\nabla \Theta_d \cdot \Gamma \leq -c_0$ with c_0 fixed by lower bounds on the drag coefficient and the speed, since the drag is strictly proportional to the density. The budget, the pass count and the bound on the time spent in the layer therefore hold for entry as they stand. The remaining hypotheses are conditions on the corridor. Normalising the gravity term to -1 in Eq. (2.2) requires $g - V^2/r$ bounded away from zero, so the corridor is bounded away from circular speed. The bank bounded away from ninety degrees gives $c \geq c_{\min} > 0$ in Assumption 2.2, and the last clause of Assumption 2.1 — that $1/c$ lie in the range of the profile — is the requirement that a glide altitude exist for every admissible bank. Assumption 2.2 also needs the corridor bounded away from the geographic poles, where the spherical chart is singular.

7.2 A second monotone quantity, and what it would cost

The application supplies a second quantity of the same kind as Assumption 2.3 but carrying a different power of the density: the integrated heat load, whose rate goes as the square root of the density where the drag power goes as the density itself. The two cut in opposite directions, and that is what makes the pair informative. For a fixed value of the first integral, concentrating the density *lowers* the second, so the first bounds how many deep excursions there can be while the second bounds how long the motion may linger at moderate density — which is a bound on the glide, and so on the number of fast orbits, that Assumption 2.3 alone does not give.

Using it is not free, and the cost is worth stating because it falls exactly on the dimension. To bound $\int \sqrt{m_\theta^\varepsilon} d\tau$ the slow equation must carry a term in $\sqrt{m_\theta^\varepsilon(z)}$ as well as one in $m_\theta^\varepsilon(z)$, since Γ does not depend on z and no condition on it alone can produce that power. And then the identity Eq. (3.3) has no counterpart:

the orbit average of $\sqrt{m_\theta^\varepsilon}$ is a function of the amplitude, not a constant, so the averaged slow motion ceases to be amplitude-free and the glide mode of Definition 4.1 must carry W as a coordinate. That is one dimension in exchange for a bound on the glide, and which way the exchange is worth making is a question for the application rather than for the model.

7.3 Why the band is not empty

No hypothesis of Theorem 4.2 fails for entry. What fails is that the two thresholds of Eq. (3.11) are not both small at the physical ε , and the reason is quantitative. A pass entered at a flight-path angle γ_e crosses a scale height in a time of order ε/γ_e on the slow time scale and levels out under a load of order γ_e^2/ε relative to gravity, so the entry angle enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman–Loh variables [18, 19]. For the Earth $\sqrt{\varepsilon} \approx 0.034$, about two degrees, and entry angles are a few degrees: a pass is therefore neither short compared with the arc that follows it nor distinct in scale from the oscillation about the glide. In the language of Eq. (3.11), no threshold W_0 makes ϑ_1 and ϑ_2 small together: the amplitude at which the dip becomes brief is the amplitude at which the period becomes long. So Theorem 4.2 holds, but its residual is not small, and Remark 4.4 rather than Eq. (4.4) is what applies. That costs no term in the residual; it costs dimension, over a total duration bounded by Eq. (3.4).

8 Plan of work

Theorem 4.2 is Theorem C for one model, and it is discharged. Five obligations remain, and we state each with the obstruction it has to overcome. The first and fourth are the analysis; the second and third carry the model to the application; the fifth is a question about evaluating a bound rather than about proving one. The first is nearer than it was: the profile count Eq. (3.12) it needs is proved, and what is open is the decomposition that count would organise.

Objective 1. Theorem A: the profile decomposition, with a constant

We must prove the decomposition itself: that every sequence of trajectories with $\varepsilon_n \rightarrow 0$ admits, along a subsequence, a splitting into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The count is to be read as the classical theory reads it, above a fixed size: by Eq. (3.12) the passes of impulse at least $2\sqrt{2W_0}/c_{\max}$ number at most $c_{\max}M/(2\sqrt{2W_0})$, whatever ε . The classical route iterates the dichotomy of the concentration-compactness principle [1, 20] and, in the form that resolves each profile at its own scale, extracts a pairwise orthogonal family [2, 21]; orthogonality is produced by the construction rather than assumed, and profiles at distinct times are orthogonal here with nothing to prove, since the scale of a profile is the width of a pass and its centre the time at which the pass occurs.

What we expect to add is the constant. In that theory the number of profiles above a given size is finite because the conserved quantity is finite, and the argument is by contradiction; here Assumption 2.3 supplies a Lyapunov functional whose total variation is bounded uniformly in ε , so the count is bounded *in advance* and explicitly, as Eqs. (3.12) and (5.2) already show. The objective is met by the decomposition stated with that count, with the remainder measured in the largest share of dissipation held by any window of fixed rescaled width — the norm in which a remainder of this kind is small, and in which it is not small in the conserved quantity [2, p. 136].

Objective 2. The statement at a fixed ε , and a smaller selection class

Theorem 4.2 holds for entry, but with a residual that Section 7 shows is not small at the physical ε . We must prove the statement Remark 4.4 describes: the orbits near the threshold carried by the unapproximated field, the reduced problem acquiring the fast pair over a total duration bounded by Eq. (3.4), and the residual gaining no new term. The enclosure of an intermediate orbit is a propagation over a bounded time rather than an asymptotic estimate, so what has to be controlled is its growth, not its order; the obligation is a bound on that growth which is explicit in the interval widths.

The same objective carries the one refinement that would most sharpen the result. In Assumption 2.2 the selection $\theta(\cdot)$ is an arbitrary measurable function: its range is bounded and its variation is not. The atmosphere is smooth in space, mass is lost monotonically, and coefficient errors are correlated, so the admissible class is far smaller than an arbitrary selection, and a bound proved over the smaller class is tighter at no cost to validity. We must state that class — bounded variation in altitude, monotonicity where it holds — and carry it through Lemmas 3.9 and 3.10. The objective is met by the fixed- ε theorem with the enclosure growth explicit, over the restricted class.

Objective 3. The remainder: motions that do not concentrate

A profile decomposition separates what concentrates from what does not, and the second part is the remainder. The entry problem has motions of that kind which Eqs. (2.1) to (2.3) does not carry: the attitude and the elastic modes oscillate for as long as there is air, and nothing about them concentrates, so in the decomposition they are not profiles at all. The wave-equation theory is careful to say that its remainder is not small in the conserved quantity, since it contains every motion that concentrating profiles cannot represent [2, p. 136], and the question is therefore in what sense the remainder *is* small here.

The classical answer is averaging [22], and for *several* phases it does not reach a bound. (Over a single orbit it does, and Lemma 4.5 uses it; the difficulty is entirely in the passage to more than one phase.) An averaging theorem gives an error to an order; with several phases it holds only away from resonance, while the attitude frequency grows with the square root of the dynamic pressure and so rises from zero at the start of every pass; and the results that carry a solution through a resonance hold outside a set of initial conditions of small measure, which an outer statement about a whole solution set cannot discard. We propose a Lyapunov functional in place of an averaging theorem — to bound what these motions *do* rather than to resolve what they are. They act on the translational motion only through the aerodynamic force, so it is the force that must be enclosed, and the enclosure may then be carried by the selection θ at no cost in dimension. Two facts are to be established. Where there is no air there is no torque, so the angular rates are bounded by conservation and the attitude has no aerodynamic effect in any case. Where there is air, the attitude damping rate is proportional to the product of density and speed, while the energy identity fixes the integral of density against the cube of speed, so the budget that bounds the dissipation also bounds the integrated damping from below — the same air does both. The objective is met by a bound on the aerodynamic force error, explicit in the dimensionless groups of the class and uniform over the corridor, with the elastic modes included or enclosed as a widened selection set.

Objective 4. The measure-level companion and the certificate

We must show that the occupation measures of Section 6 converge to measures satisfying the Liouville equation of the limit system, with the mass crossing each pass carried by a measure on the graph of J , uniformly over the admissible family, and exhibit the dual pair (Φ, Ψ) non-decreasing along the arc flow and across the reset. This is what makes Theorem 4.2 a statement about the limit of the family rather than a comparison of two trajectories, and it is where weak-* compactness and concentration-compactness in the form that resolves each profile at its own scale [1, 20] are expected to be needed — for the structure, with the constants coming from the functionals of Assumption 2.3 as in objective 1.

Objective 5. Validated evaluation of the certificate

The guarantee is to be verified in validated arithmetic rather than produced by a numerical solver, and this is the obligation set aside in [Section 1.1](#). We must show that the closed-form constants, the enclosures of the passes they do not cover, and the reachable set of the limit system can be evaluated over the whole corridor with every rounding error accounted for. Two losses have to be managed and neither is removed by refining a step: repeated occurrences of a variable are treated as independent, and the image of a box under a flow is not a box, so re-enclosing it compounds [\[3, Chs. 4–5, 10\]](#). What controls the second is the representation of the set [\[4, 15, 23\]](#). The objective is met when every inequality the certificate asserts is discharged by a computation returning either a proof or an explicit failure to decide at the working precision.

8.1 Sequencing

Three phases rather than five parallel tracks. The first extends [Theorem 4.2](#) to the planar entry problem, where the correspondence above is exact and [Assumptions 2.1](#) and [2.3](#) are verified rather than assumed; the question of which of the three terms of [Eq. \(4.3\)](#) is the real obstruction, raised in [Remark 4.3](#), belongs to it, and so does [objective 1](#), since the profile count [Eq. \(3.12\)](#) is already in hand and what remains is to state the decomposition it bounds. The second is [objective 2](#), in which the threshold of [Eq. \(3.11\)](#) and the selection class are the two dials on the same statement. The third is [objective 3](#), and it should be reached early enough that a negative result is still recoverable; [objective 4](#) runs alongside it, needing only that the limit system be settled, and [objective 5](#) follows whatever the first phase settles, since it is a question about evaluating whichever bound that phase leaves. If the Lyapunov route of [objective 3](#) does not close, the retreat is a corridor on which the frequency bands are ordered, with the crossings at the edges of each pass left to the enclosure of the bubble, which integrates the coupled field and needs no separation; that weakens the residual without invalidating the decomposition.

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